

NEELAY RANJAN

Generative Modeling for Safety-Critical, Data-Scarce Domains

@ neelay.ranjan@outlook.com +1 (773) 668-0069 in neelayranjan NeelayRanjan neelayranjan.dev ID 0009-0008-9482-0160

Generative-modeling researcher building diffusion and language-model systems for safety-critical domains where data is scarce and failure is expensive: first-author work on label-efficient vessel segmentation that beats SAM with 7 to 22 labels, and hazard-aware routing and air traffic control speech pipelines at NASA Ames.

RESEARCH & WORK EXPERIENCE

NASA Ames Research Center, Aeronautics Directorate Moffett Field / Mountain View, CA
GenAI Applied Research Intern, SHIFT Aug 2026 – Present

- Fine-tuned Qwen3-Omni-30B with QLoRA for air traffic control (ATC) speech-to-text on ~1.5 h of real audio (~900 utterances): **17% WER vs. 20% for the existing Whisper ASR** and 30% zero-shot (5-fold CV).
- Replaced LLM parsing of transcripts into the DTI maneuver ontology with Kev, a typed decision model that cannot invent fields: **value-match accuracy 40% → 80%** over the best rules-based and LLM attempts, at ~90% lower latency and ~400× lower serving cost (vs. Qwen3 27B).
- Building a text-to-speech synthetic ATC corpus to expand the dataset ~20× (5:95 real:synthetic); outputs feed a maneuver database matched to historical and live FAA airspace records for safety research.

GenAI Applied Research Intern, SLAAC May – Aug 2026

- Built hazard-aware flight rerouting with diffusion models learned from FAA radar tracks, with mid-sampling guidance that steers around special-use airspace at inference with no retraining: **99% of reroutes clear the 25 nm hazard buffer at a median +10 nm (1.1%) added distance**, within 1.2% of the geometric optimum.
- Designed a flight-plan LLM with a custom tokenizer that synthesizes a full day of ~**44,000 FAA-managed flights** matched to historical traffic density, producing simulation-ready traffic days for the software NASA and the FAA use to evaluate future air traffic management (ATM) strategies.

Regenstrief Institute Indianapolis, IN
Applied Vision-AI Researcher Feb 2024 – Present

- Showed an x0 flipped-objective diffusion segmenter outperforms SAM (ViT-B) on benchmark angiograms in ultra-low-data regimes: **88.2% Dice with just 16 labels**; its pre-labels bootstrap surgeon labeling campaigns, **cutting correction time by 75%**.
- Built a modular SAM-x0 tiled refinement pipeline (SAM prior + partial-resolution x0 diffusion refiner) for real-world, region-agnostic angiograms at native 1024 resolution: **70% Dice with 22 labeled images**, beating both SAM (46%) and raw x0 (32%), at a fraction of full-resolution compute.
- Extended the pipeline to cerebral moyamoya MRA, segmenting vasculature from **only 7 labeled images** via patient-level cross-validation and tiling augmentation: ~55% Dice vs. SAM’s ~20%.

SELECTED PUBLICATIONS

- N. Ranjan et al. “Bootstrapping surgeon labeling campaigns with x0-diffusion: label-efficient vessel segmentation of catheter-based angiograms.” *First author, in preparation.*
- N. Ranjan et al. “Modular SAM-prior diffusion refinement for real-world angiogram and seven-shot moyamoya MRA vessel segmentation.” *First author, in preparation.*
- Co-author, “Helical Antenna for Electromagnetic Field Stimulation in Alzheimer’s Disease Therapy.” *Presented at IEEE MWSCAS 2026; led the PCB design team.*

PROJECTS & APPLIED RESEARCH

Energy-Based Model for Chess → Solar-Powered Inference Device 2025

- Trained an energy-based model that scores and ranks moves from a learned energy landscape over game states, reaching ~**1900–2200 estimated Elo** against a ladder of rated Stockfish depths.
- Embedded the EBM on a Raspberry Pi Zero 2 W (~1 s / move) as a standalone chess engine **powered solely by onboard solar & supercapacitors**; quantization tests showed FP32 beat INT8 on ARM (no native INT8 kernels).

LLM Design Team Lead – Aircraft Maintenance Assistant, V2X Aug 2024 – May 2025

- Led design and fine-tuning of a maintenance-assistant LLM for V2X (Raytheon divestiture); its two-stage RAG pipeline (summary-level source selection → section-level pinpointing with direct quoting) **cut hallucinations from ~40% to ~5%**.

KEY RESULTS

20% → 17% WER
QLoRA ATC speech-to-text, 5-fold CV

40% → 80% value match
Kev typed parsing, 90% lower latency

99% hazard clearance
Diffusion reroutes at +1.1% distance

88.2% Dice at 16 labels
x0 segmenter vs. SAM on angiograms

75% faster correction
x0 pre-labels speed surgeon labeling

70% Dice at 1024 px
Real-world angiograms (SAM 46%)

7 training images
Moyamoya MRA at ~55% (SAM ~20%)

V2X **40% → 5% hallucinations**
Two-stage RAG, maintenance LLM

EDUCATION

Purdue University, College of Engineering
B.S. in Artificial Intelligence
 Expected May 2027 Indianapolis, IN
Concentration: Intelligent Control & Systems
Minor: Mathematics
GPA: 3.68 / 4.0 · Dean’s List
John Martinson Honors College
CITI: Biomedical Research, RCR, GCP

TECHNICAL SKILLS

Generative modeling

foundation-model adaptation (SAM)
diffusion (x0 / v-prediction) transformers
energy-based models LLM fine-tuning & RAG
safety-critical AI interpretability
multimodal pipelines

ML engineering

Python C/C++ CUDA PyTorch Git
medical image segmentation Docker W&B
Linux / cluster administration quantization
transfer learning Slurm / H100

Hardware / edge

PCB design ultra-low-power inference
embedded ML